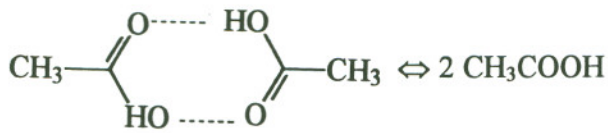
PHYSICAL CHEMISTRY CUMULATIVE EXAM
December 3, 1999

At lower temperatures, acetic acid in the gas phase consists mainly of the dimer. The dimer dissociates according to



To determine the equilibrium constant and the enthalpy of dissociation of the dimer, a container of volume 21.5 cm^3 is filled with acetic acid vapor at 437 K and 764 Torr pressure. After cooling, the acid in the container is analyzed and its mass is found 0.052 g. The experiment is repeated at the same pressure but at 471 K, and the mass of the acid is 0.038 g. Calculate the equilibrium constant for dissociation at these two temperatures,

$$K_P = (P_{acid}/p^o)^2/(P_{dimer}/p^o)$$

assuming that both acetic acid and its dimer behave like perfect gases. Also calculate the enthalpy of dimerization from the van't Hoff equation, $d \ln K/ d(1/T) = - \Delta_r H^o/R$ in kJ/mol. The standard pressure p^o = 750 Torr, the gas constant expressed with liter and Torr is R=62.364 $LTorrK^{-1}mol^{-1}$, in SI units R=8.3145 J/(K mol), the molecular mass of acetic acid is 60.05 g/mol